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Inventor(s)	Sakamoto; Ryota et al.

Driving apparatus and driving method for driving lens with corrected tilt

Abstract

Provided is a driving apparatus including: a calculation unit which calculates a driving amount of an optical element based on a detection position and target position of the optical element; a correction unit which generates a correction driving amount obtained by correcting the driving amount based on the driving amount, a parameter corresponding to a tilt of the optical element, and either the detection position or the target position, and outputs the driving amount and the correction driving amount; a first driving unit which applies a first driving force corresponding to either one of the driving amount or the correction driving amount to a first driving source out of driving sources for moving the optical element in a predetermined direction; and a second driving unit which applies a second driving force corresponding to another one of the driving amount or the correction driving amount to a second driving source.

Inventors:	Sakamoto; Ryota (Tokyo, JP), Yokobori; Takumi (Tokyo, JP)
Applicant:	Asahi Kasei Microdevices Corporation (Tokyo, JP)
Family ID:	1000008779859
Assignee:	Asahi Kasei Microdevices Corporation (Tokyo, JP)
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Primary Examiner: Haliyur; Padma

Background/Summary

(1) The contents of the following Japanese patent application(s) are incorporated herein by reference: NO. 2022-007064 filed in JP on Jan. 20, 2022 NO. 2022-177005 filed in JP on Nov. 4, 2022

BACKGROUND

1. Technical Field

(2) The present invention relates to a driving apparatus and a driving method.

2. Related Art

(3) Patent Document 1 describes that “the controller applies a predetermined offset force to a driving force generated in one driving source, and controls a driving force of the other driving source so as to compensate for an imbalance of the driving forces caused by the offset force”.

PRIOR ART DOCUMENT

Patent Document

(4) Patent Document 1: Japanese Patent Application Publication No. 2018-045484

SUMMARY

(5) A first aspect of the present invention provides a driving apparatus. The driving apparatus may include a calculation unit configured to calculate a driving amount of an optical element based on a detection position of the optical element and a target position of the optical element. The driving

apparatus may include a correction unit to which the driving amount, the detection position, and the target position are input, the correction unit being configured to generate a correction driving amount obtained by correcting the driving amount based on the driving amount, a parameter corresponding to a tilt of the optical element, and either the detection position or the target position, and output the driving amount and the correction driving amount. The driving apparatus may include a first driving unit configured to apply a first driving force corresponding to either one of the driving amount or the correction driving amount to a first driving source out of a plurality of driving sources for moving the optical element in a predetermined direction. The driving apparatus may include a second driving unit configured to apply a second driving force corresponding to another one of the driving amount or the correction driving amount to a second driving source out of the plurality of driving sources.

(6) The correction unit may be configured to correct the driving amount based on a function of the detection position.

(7) When a predetermined first coefficient is set as a “coefficient 1” and a predetermined second coefficient is set as a “coefficient 2”, the correction unit may be configured to correct a driving current corresponding to the driving amount calculated by the calculation unit using an expression “corrected driving current=driving current×coefficient 1+driving current×detection position×coefficient 2”.

(8) The correction unit may be configured to correct the driving amount using a correction table predefined based on the detection position and the driving amount.

(9) The correction unit may be configured to correct the driving amount based on a function of the target position.

(10) When a predetermined first coefficient is set as a “coefficient 1” and a predetermined second coefficient is set as a “coefficient 2”, the correction unit may be configured to correct a driving current corresponding to the driving amount calculated by the calculation unit using an expression “corrected driving current=driving current×coefficient 1+driving current×target position×coefficient 2”.

(11) The correction unit may be configured to correct the driving amount using a correction table predefined based on the target position and the driving amount.

(12) The correction unit may be configured to be capable of switching whether to use either the detection position or the target position for correcting the driving amount.

(13) The correction unit may include a switching unit configured to make, according to characteristics of the first driving source and the second driving source, a switch between a correction that uses either the detection position or the target position and a correction that does not use the detection position and the target position.

(14) The correction unit may be configured to select, based on the parameter corresponding to the tilt of the optical element, which of the plurality of driving sources is to be used as a reference for correcting the driving amount.

(15) The parameter corresponding to the tilt may indicate a tilt of the optical element that is detected when a maximum current is caused to flow into each of the plurality of driving sources.

(16) The calculation unit may be configured to calculate the driving amount by feedback control that uses the detection position and the target position.

(17) A second aspect of the present invention provides a driving method. The driving method may be a method of driving a plurality of driving sources including a first driving source and a second driving source. The driving method may include detecting a tilt of an optical element. The driving method may include generating a parameter corresponding to the tilt of the optical element. The driving method may include detecting a position of the optical element and outputting a detection position. The driving method may include calculating a driving amount of the optical element based on the detection position of the optical element and a target position of the optical element. The driving method may include generating a correction driving amount obtained by correcting the

driving amount based on the driving amount, the parameter corresponding to the tilt of the optical element, and either the detection position or the target position. The driving method may include selecting, based on the parameter corresponding to the tilt of the optical element, which of the first driving source and the second driving source is to be driven by a driving force corresponding to the correction driving amount.

(18) The summary clause does not necessarily describe all necessary features of the embodiments of the present invention. The present invention may also be a sub-combination of the features described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows an example of a block diagram of a camera module **10** that may include a driving apparatus **100** according to the present embodiment.

(2) FIG. 2 shows an example of a flow in which the driving apparatus **100** according to the present embodiment executes tilt correction processing.

(3) FIG. 3 shows an example of a block diagram of the camera module **10** that may include the driving apparatus **100** according to a second embodiment.

(4) FIG. 4 shows an example of a block diagram of the camera module **10** that may include the driving apparatus **100** according to a third embodiment.

(5) FIG. 5 shows an example of a block diagram of the camera module **10** that may include the driving apparatus **100** according to a fourth embodiment.

(6) FIG. 6 shows an example of a block diagram of a correction unit **130** in the driving apparatus **100** according to a fifth embodiment.

(7) FIG. 7 shows an example of a block diagram of the camera module **10** that may include the driving apparatus **100** according to a sixth embodiment.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(8) Hereinafter, embodiments of the present invention will be described, but the embodiments do not limit the invention according to the claims. In addition, not all combinations of features described in the embodiments are essential to the solution of the invention.

(9) FIG. 1 shows an example of a block diagram of a camera module **10** that may include a driving apparatus **100** according to the present embodiment. It is to be noted that these blocks are functional blocks that are functionally separated, and do not necessarily need to match with actual device configurations. In other words, in the present figure, a unit shown as one block does not necessarily need to be configured by one device. Also, in the present figure, units shown as separate blocks do not necessarily need to be configured by separate devices. The same also applies to other figures.

(10) Further, hereinafter, the camera module **10** will be described as an example, but the present invention is not limited to this. Portable electronic equipment or a position control system having functions similar to those of the camera module **10** to be described below may also be provided. Examples of such equipment or system include a cellular phone, a smartphone, a tablet terminal, a PDA, a portable computer, a laptop, a notebook personal computer, a built-in or external system that controls a position of a lens, and the like. Furthermore, although a lens is taken as an example of an optical element in the descriptions, the optical element may not only be the lens but may also be an image sensor.

(11) The camera module **10** may be capable of executing processing such as Autofocus (AF) or Zoom. In the present embodiment, the camera module **10** executes tilt correction processing for correcting a tilt of a lens in cooperation with the above-described processing or independently from the above-described processing.

- (12) The camera module **10** includes an object **20**, a plurality of driving sources **50**, a processor **70**, and a driving apparatus **100**.
- (13) The object **20** is a device to be driven. Hereinafter, a case where the object **20** is a lens tube will be described as an example. In the object **20**, a lens **30** and a first magnet **40_1** and second magnet **40_2** (collectively referred to as magnets **40**) are provided integrally.
- (14) The lens **30** is an optical element for refracting and converging light. When executing AF processing or Zoom processing, the camera module **10** causes such a lens **30** to move linearly along an optical axis direction so as to perform focusing or zoom in/zoom out.
- (15) The magnets **40** are permanent magnets. In the present embodiment, a case where the first magnet **40_1** and the second magnet **40_2** are provided as the magnets **40** is described as an example. For example, the first magnet **40_1** and the second magnet **40_2** may be provided at positions opposing each other with the lens **30** interposed therebetween along the optical axis direction of the lens **30**.
- (16) The plurality of driving sources **50** are sources of a thrust force for causing the lens **30** to move in a predetermined direction, herein, the optical axis direction of the lens **30**. In the present embodiment, a case where the plurality of driving sources **50** include a first driving source **50_1** and a second driving source **50_2** is described as an example. Each of the plurality of driving sources **50** may include a coil, for example. The first driving source **50_1** may include, in the vicinity of the first magnet **40_1**, a coil wound along the optical axis direction of the lens **30**. In addition, the second driving source **50_2** may include, in the vicinity of the second magnet **40_2**, a coil wound along the optical axis direction of the lens **30**. In such a plurality of driving sources **50**, for example, when a driving current or a driving voltage is supplied and a driving force is applied, a magnetic force is generated between the first driving source **50_1** and the first magnet **40_1** and between the second driving source **50_2** and the second magnet **40_2**, and thus this magnetic force becomes a thrust force so that a position of the lens **30** can be moved.
- (17) The processor **70** is a processing apparatus that assumes control of the camera module **10**. For example, the processor **70** may be communicably connected to the driving apparatus **100** via serial communication, parallel communication, a network, wireless communication, or the like. As an example, I2C (Inter-Integrated Circuit) may be used for such communication. The processor **70** may supply a target position signal indicating a target position of the lens **30** to the driving apparatus **100**. In addition, the processor **70** may supply tilt information representing a tilt of the lens **30** to the driving apparatus **100**.
- (18) The driving apparatus **100** acquires the target position signal and the tilt information from the processor **70**. Then, the driving apparatus **100** calculates a driving amount of the lens **30** based on the detection position and target position of the lens **30**, and applies a driving force corresponding to the calculation result to the plurality of driving sources **50**. At this time, the driving apparatus **100** according to the present embodiment corrects the calculated driving amount based on the position of the lens **30** and the tilt of the lens **30**.
- (19) The driving apparatus **100** includes a position sensor **110**, a calculation unit **120**, a correction unit **130**, and a plurality of driving units **140**. It is to be noted that in the present figure, a case where the plurality of driving units **140** include a first driving unit **140_1** and a second driving unit **140_2** is shown as an example.
- (20) The position sensor **110** detects a position of the lens **30**. For example, the position sensor **110** may be a magnetic sensor, and may detect the position of the lens **30** by detecting a magnetic field generated by the magnets **40** provided integrally with the lens **30**. As an example, such a magnetic sensor may be a Hall element to which a Hall effect is applied and which senses a variation in an external magnetic field from a generated electromotive force, such as a silicon Hall element and a compound Hall element. However, the present invention is not limited to this. The magnetic sensor may be various sensors capable of detecting a magnetic field, such as a spin-valve type magnetoresistance effect element (GMR element, TMR element, or the like) in which a resistance

varies according to a variation in the external magnetic field, or may be a combination of these various sensors. Alternatively, the position sensor **110** may be configured by a sensor element group constituted of a plurality of sensor elements. Further, the position sensor **110** may amplify a detected amplitude voltage or current value to be 1 time or more, or may convert it into a digital value by A/D conversion. The position sensor **110** supplies a detection position signal indicating the detection position obtained by detecting the position of the lens **30** to the calculation unit **120**. Furthermore, in the present embodiment, the position sensor **110** also supplies the detection position signal to the correction unit **130** in addition to the calculation unit **120**. It is to be noted that although the present figure shows a case where the position sensor **110** is built-in in the driving apparatus **100** as an example, the position sensor **110** may be configured separately from the driving apparatus **100** and be externally attached to the driving apparatus **100**.

(21) The calculation unit **120** calculates a driving amount of the optical element based on the detection position of the optical element (lens **30**) and the target position of the optical element. At this time, the calculation unit **120** may execute a PID calculation. Herein, PID is one type of feedback control and is control in which control of an input value is performed using three elements including a deviation between an output value and a target value, an integral thereof, and a differential thereof. There is proportional control (P control) as basic feedback control. This is to control the input value as a linear function of the deviation between the output value and the target value. An operation of varying the input value in proportion to this deviation is called a proportional operation or a P operation (P is an abbreviation of Proportional). That is, when a state with the deviation continues for a long time, the P operation serves to increase the variation in the input value accordingly to bring it close to the target value. Moreover, an operation of varying the input value in proportion to an integral of this deviation is called an integral operation or an I operation (I is an abbreviation of Integral). Control in which the proportional operation and the integral operation are combined in this manner is called PI control. Further, an operation of varying the input value in proportion to a differential of this deviation is called a differential operation or a D operation (D is an abbreviation of Derivative or Differential). Such control in which the proportional operation, the integral operation, and the differential operation are combined is called PID. That is, the calculation unit **120** executes the PID calculation based on the detection position of the lens **30** detected by the position sensor **110** and the target position of the lens **30** designated by the processor **70**, to thus calculate the driving amount of the lens **30**. In other words, the calculation unit **120** executes closed loop feedback control so as to move the lens **30** to a target position based on a result of detecting the position of the lens **30**. In this manner, for example, the calculation unit **120** can calculate the driving amount by the feedback control that uses the detection position and the target position. The calculation unit **120** supplies the calculation result, that is, information representing the driving amount of the lens **30**, to the correction unit **130**.

(22) The correction unit **130** is input with the driving amount, the detection position, and the target position, generates a correction driving amount obtained by correcting the driving amount based on the driving amount, a parameter corresponding to a tilt of the optical element (lens **30**), and either the detection position or the target position, and outputs the driving amount and the correction driving amount. In the present embodiment, the correction unit **130** corrects the driving amount of the lens **30** based on the detection position of the lens **30** and the tilt of the lens **30**. The correction unit **130** outputs the driving amount and the correction driving amount obtained by correcting the driving amount to the plurality of driving units **140**. Herein, regarding the parameter corresponding to the tilt of the lens **30**, tilt information at each lens position may be acquired in advance in an actuator inspection process or the like, and during an actual operation, the correction unit **130** may perform the correction based on the tilt information that has been acquired in advance.

(23) The first driving unit **140_1** applies a first driving force corresponding to either one of the driving amount or the correction driving amount to the first driving source **50_1** out of the plurality of driving sources **50** for moving the lens **30** in a predetermined direction. More specifically, the

first driving unit **140_1** supplies a driving current or driving voltage corresponding to either one of the driving amount calculated by the calculation unit **120** or the correction driving amount generated by the correction unit **130** to the first driving source **50_1**, to thus apply the first driving force to the first driving source **50_1**.

(24) The second driving unit **140_2** applies a second driving force corresponding to the other one of the driving amount or the correction driving amount to the second driving source **50_2** out of the plurality of driving sources **50**. More specifically, the second driving unit **140_2** supplies a driving current or driving voltage corresponding to the other one of the driving amount calculated by the calculation unit **120** or the correction driving amount generated by the correction unit **130** to the second driving source **50_2**, to thus apply the second driving force to the second driving source **50_2**.

(25) In the plurality of driving sources **50**, when such a driving current or driving voltage is supplied, a magnetic force is generated between the first driving source **50_1** and the first magnet **40_1** and between the second driving source **50_2** and the second magnet **40_2**, and thus this magnetic force becomes a thrust force so that a position of the lens **30** can be moved. Then, the driving apparatus **100** detects a position to which the lens **30** has moved by the position sensor **110** and feeds back the result to the calculation unit **120**, to thereby move the lens **30** to a target position. At this time, according to the driving apparatus **100** of the present embodiment, since the correction unit **130** corrects a driving capability of either one of the first driving source **50_1** or the second driving source **50_2**, the lens **30** can be moved without tilting with respect to the optical axis.

(26) It is to be noted that the correction unit **130** may be capable of selecting which of the plurality of driving sources **50** is to be used as a reference for correcting the driving amount. This will be described in detail by using a flow.

(27) FIG. 2 shows an example of a flow in which the driving apparatus **100** according to the present embodiment executes tilt correction processing. In Step S210, the driving apparatus **100** causes a maximum current to flow into the plurality of driving sources **50**. For example, the plurality of driving units **140** supply a driving current to each of the plurality of driving sources **50** such that the driving current becomes a maximum current.

(28) In Step S220, the driving apparatus **100** acquires tilt information representing a tilt of the lens **30**. For example, the correction unit **130** acquires tilt information representing the tilt of the lens **30** from the processor **70**. At this time, the parameter corresponding to the tilt of the optical element (lens **30**) indicates the tilt of the lens **30** detected when a maximum current is caused to flow into each of the plurality of driving sources **50**. It is to be noted that Step S210 and Step S220 may be performed as a part of the actuator inspection process.

(29) In Step S230, the driving apparatus **100** confirms a driving capability. For example, the correction unit **130** confirms which of the first driving source **50_1** and the second driving source **50_2** the lens **30** is tilted toward based on the tilt information acquired in Step S220. In general, the driving capability of the driving source **50** depends on mechanical characteristics of the actuator. At this time, since the lens **30** moves more toward the driving source having a higher driving capability, the lens **30** tilts with respect to the optical axis when there is a difference in the driving capabilities between the plurality of driving sources **50**. Herein, it is difficult to correct the lower driving capability so that it matches with the higher driving capability. In this regard, the driving apparatus **100** according to the present embodiment corrects the higher driving capability so that it matches with the lower driving capability. Therefore, the correction unit **130** may select which of the plurality of driving sources **50** is to be used as a reference for correcting the driving amount based on the parameter corresponding to the tilt of the optical element (lens **30**).

(30) When determined that the driving capability of the first driving source **50_1** is lower based on the tilt of the lens **30**, the driving apparatus **100** advances the processing to Step S240. On the other hand, when determined that the driving capability of the second driving source **50_2** is lower based

on the tilt of the lens **30**, the driving apparatus **100** advances the processing to Step S250.

(31) In Step S240, the driving apparatus **100** corrects the driving amount calculated by the calculation unit **120** in order to correct the driving capability of the second driving source **50_2** using the first driving source **50_1** as a reference.

(32) In the present embodiment, the correction unit **130** corrects the driving amount calculated by the calculation unit **120** based on the detection position of the lens **30** and the tilt of the lens **30**. As an example, when a predetermined first coefficient is set as a “coefficient 1” and a predetermined second coefficient is set as a “coefficient 2”, the correction unit **130** corrects the driving current corresponding to the driving amount calculated by the calculation unit **120** using, for example, the following expression “corrected driving current=driving current×coefficient 1+driving current×detection position×coefficient 2”. That is, the correction unit **130** corrects the driving amount based on a function of the detection position of the lens **30**. More specifically, the correction unit **130** corrects the driving amount by a sum of: a function that uses the driving current corresponding to the driving amount calculated by the calculation unit **120** and the predetermined coefficient 1 as variables; and a function that uses the driving current corresponding to the driving amount calculated by the calculation unit **120**, the detection position of the lens **30**, and the predetermined coefficient 2 as variables. Herein, at least one of the “coefficient 1” or the “coefficient 2” may be predefined such that it becomes a numerical value corresponding to a magnitude of the tilt of the lens **30**.

(33) It is to be noted that although the case where the correction unit **130** corrects the driving amount based on the function of the detection position of the lens **30** is described as an example in the descriptions above, the present invention is not limited to this. The correction unit **130** may correct the driving amount using a correction table predefined based on the detection position of the lens **30** and the driving amount.

(34) Then, the correction unit **130** outputs the correction driving amount obtained by correcting the driving amount to the second driving unit **140_2**. Accordingly, the second driving unit **140_2** applies a second driving force corresponding to the correction driving amount to the second driving source **50_2** out of the plurality of driving sources **50**. More specifically, the second driving unit **140_2** supplies a corrected driving current to the second driving source **50_2**, to thus apply the second driving force to the second driving source **50_2**. On the other hand, the correction unit **130** outputs the driving amount not yet subjected to the correction, that is, the driving amount calculated by the calculation unit **120**, to the first driving unit **140_1**. Accordingly, the first driving unit **140_1** applies a first driving force corresponding to the driving amount not yet subjected to the correction to the first driving source **50_1** out of the plurality of driving sources **50**. More specifically, the first driving unit **140_1** supplies a driving current not yet subjected to the correction to the first driving source **50_1**, to thus apply the first driving force to the first driving source **50_1**. In this manner, the driving apparatus **100** corrects the driving capability of the second driving source **50_2** using the first driving source **50_1** having a lowest driving capability as a reference.

(35) On the other hand, in Step S250, the driving apparatus **100** corrects the driving amount calculated by the calculation unit **120** in order to correct the driving capability of the first driving source **50_1** using the second driving source **50_2** as a reference. It is to be noted that since the correction of the driving amount may be similar to that of Step S240, detailed descriptions will be omitted herein.

(36) Then, the correction unit **130** supplies the correction driving amount obtained by correcting the driving amount to the first driving unit **140_1**. Accordingly, the first driving unit **140_1** applies the first driving force corresponding to the corrected driving amount to the first driving source **50_1** out of the plurality of driving sources **50**. More specifically, the first driving unit **140_1** supplies the corrected driving current to the first driving source **50_1**, to thus apply the first driving force to the first driving source **50_1**. On the other hand, the correction unit **130** outputs the driving

amount not yet subjected to the correction, that is, the driving amount calculated by the calculation unit **120**, to the second driving unit **140_2**. Accordingly, the second driving unit **140_2** applies the second driving force corresponding to the driving amount not yet subjected to the correction to the second driving source **50_2** out of the plurality of driving sources **50**. More specifically, the second driving unit **140_2** supplies the driving current not yet subjected to the correction to the second driving source **50_2**, to thus apply the second driving force to the second driving source **50_2**. In this manner, the driving apparatus **100** corrects the driving capability of the first driving source **50_1** using the second driving source **50_2** having a lowest driving capability as a reference.

(37) In recent years, for improving a driving force of a lens due to an increase in a size and weight of the lens, a plurality of driving sources may be used for moving the lens. However, in such a case, when the driving capabilities vary between the plurality of driving sources, a situation where the lens is tilted has occurred.

(38) Patent Document 1 describes a technique of compensating an imbalance of driving capabilities of driving sources by adding an offset to a driving amount. However, Patent Document 1 does not describe anything regarding a tilt correction of an object due to the imbalance of the driving capabilities. Further, when the driving current is supplied as it is to the plurality of driving units without correcting the calculated driving amount, the lens may tilt when moving the lens in the optical axis direction due to the variation in the driving capabilities of the plurality of driving sources. Moreover, even if the driving amount is corrected without considering the position of the lens as in corrected driving current=driving current×coefficient 1+coefficient 2 or the like, in a case where the driving capabilities of the plurality of driving sources vary depending on the position of the lens, there has been a problem that, even when the tilt can be corrected at the lens position x1, the tilt cannot be fully corrected at a different position x2.

(39) In contrast, the driving apparatus **100** according to the present embodiment corrects the driving amount based on the detection position of the lens **30** and the tilt of the lens **30**. Then, the driving apparatus **100** according to the present embodiment applies the first driving force corresponding to either one of the driving amount not yet subjected to the correction or the corrected driving amount to the first driving source, and applies the second driving force corresponding to the other one of the driving amount not yet subjected to the correction or the corrected driving amount to the second driving source. Thus, according to the driving apparatus **100** of the present embodiment, even when the plurality of driving sources **50** are used for moving the lens in a predetermined direction, the tilt of the lens **30** due to the imbalance of the driving capabilities between the plurality of driving sources **50** can be corrected according to the position of the lens **30**.

(40) At this time, the driving apparatus **100** according to the present embodiment may correct the driving amount based on the function of the detection position. Thus, according to the driving apparatus **100** of the present embodiment, an appropriate correction calculation can be performed according to the position of the lens **30** that has actually been detected. Further, the driving apparatus **100** according to the present embodiment may correct the driving amount using a correction table predefined based on the detection position and the driving amount. Thus, according to the driving apparatus **100** of the present embodiment, a processing load of the correction calculation that is based on the detection position can be reduced.

(41) In addition, the driving apparatus **100** according to the present embodiment may select, based on the tilt of the lens **30**, which of the plurality of driving sources **50** is to be used as a reference for correcting the driving amount. At this time, the driving apparatus **100** according to the present embodiment may use, as the tilt of the lens **30**, the tilt of the lens **30** that is detected when a maximum current is caused to flow into each of the plurality of driving sources **50**. Thus, according to the driving apparatus **100** of the present embodiment, it is possible to correct, after confirming the driving capabilities of the plurality of driving sources **50** based on the tilt of the lens **30** when a maximum current is actually caused to flow, for example, the higher driving capability so that it

matches with the lower driving capability.

(42) FIG. 3 shows an example of a block diagram of the camera module **10** that may include the driving apparatus **100** according to a second embodiment. In the present figure, components having the same functions and configurations as those of FIG. 1 are denoted by the same reference numerals, and descriptions thereof will be omitted below except for different points. In the embodiment described above, the case where the driving apparatus **100** executes the tilt correction processing based on the detection position of the lens **30** and the tilt of the lens **30** has been described as an example. In the second embodiment, the driving apparatus **100** executes the tilt correction processing based on a target position of the lens **30** and the tilt of the lens **30**.

(43) In the second embodiment, the correction unit **130** acquires, instead of acquiring the detection position signal indicating the detection position of the lens **30** from the position sensor **110**, a target position signal indicating a target position of the lens **30** from the processor **70**.

(44) In addition, in the second embodiment, the correction unit **130** corrects the driving amount calculated by the calculation unit **120** based on the target position of the lens **30** and the tilt of the lens **30**. As an example, the correction unit **130** corrects the driving current corresponding to the driving amount calculated by the calculation unit **120** using, for example, the following expression “corrected driving current=driving current×coefficient 1+driving current×target position×coefficient 2”. That is, the correction unit **130** corrects the driving amount based on a function of the target position of the lens **30**. More specifically, the correction unit **130** corrects the driving amount by a sum of: a function that uses the driving current corresponding to the driving amount calculated by the calculation unit **120** and the predetermined coefficient 1 as variables; and a function that uses the driving current corresponding to the driving amount calculated by the calculation unit **120**, the target position of the lens **30**, and the predetermined coefficient 2 as variables. Herein, at least one of the “coefficient 1” or the “coefficient 2” may be predefined such that it becomes a numerical value corresponding to a magnitude of the tilt of the lens **30**.

(45) It is to be noted that although the case where the correction unit **130** corrects the driving amount based on the function of the target position of the lens **30** is described as an example in the descriptions above, the present invention is not limited to this. The correction unit **130** may correct the driving amount using a correction table predefined based on the target position of the lens **30** and the driving amount.

(46) In this manner, the driving apparatus **100** according to the second embodiment corrects the driving amount based on the function of the target position instead of the detection position. Thus, according to the driving apparatus **100** of the second embodiment, an appropriate correction calculation can be performed according to the target position of the lens **30** designated by the processor **70** instead of the position of the lens **30** that has actually been detected. Moreover, the driving apparatus **100** according to the second embodiment may correct the driving amount using a correction table predefined based on the target position and the driving amount. Thus, according to the driving apparatus **100** of the second embodiment, a processing load of the correction calculation that is based on the target position can be reduced.

(47) It is to be noted that in the descriptions above, the tilt correction processing that is based on the detection position and the tilt correction processing that is based on the target position have been described as different embodiments. However, these tilt correction processing may be used in combination. That is, the driving apparatus **100** may execute the tilt correction processing based on the detection position and target position of the lens and the tilt of the lens. Thus, according to the driving apparatus **100**, the correction calculation can be performed according to both the position of the lens **30** that has actually been detected and the target position of the lens **30** designated by the processor **70**.

(48) FIG. 4 shows an example of a block diagram of the camera module **10** that may include the driving apparatus **100** according to a third embodiment. In the present figure, components having the same functions and configurations as those of FIG. 1 are denoted by the same reference

numerals, and descriptions thereof will be omitted below except for different points. In the embodiments described above, the case where the camera module **10** causes the lens **30** to move using two driving sources has been described as an example. However, the present invention is not limited to this. The camera module **10** may cause the lens **30** to move using three or more driving sources.

(49) In the third embodiment, a case where the plurality of driving sources **50** include a third driving source **50_3** in addition to the first driving source **50_1** and the second driving source **50_2** is described as an example. Further, in association with this, the plurality of driving units **140** include a third driving unit **140_3** in addition to the first driving unit **140_1** and the second driving unit **140_2**. The third driving unit **140_3** applies a third driving force corresponding to either the driving amount or the correction driving amount to the third driving source **50_3** out of the plurality of driving sources **50**. More specifically, the third driving unit **140_3** supplies a driving current or driving voltage corresponding to either the driving amount calculated by the calculation unit **120** or the correction driving amount generated by the correction unit **130** to the third driving source **50_3**, to thus apply the third driving force to the third driving source **50_3**.

(50) In this manner, even when the plurality of driving sources **50** include three or more driving sources, the correction unit **130** may be capable of selecting which driving source is to be used as a reference for correcting the driving amount. That is, the driving apparatus **100** causes a maximum current to flow into each of the plurality of driving sources **50**, and confirms the driving capabilities of the plurality of driving sources **50**. Then, when determined that the driving capability of the first driving source **50_1** is the lowest, the correction unit **130** corrects the driving capabilities of the second driving source **50_2** and the third driving source **50_3** using the first driving source **50_1** as a reference.

(51) Therefore, in the third embodiment, the correction unit **130** requires a plurality of blocks corresponding to the plurality of driving sources **50** to be correction targets, that is, two blocks including a block for correcting the driving capability of the second driving source **50_2** and a block for correcting the driving capability of the third driving source **50_3** in the example described above. In this regard, in the third embodiment, the correction unit **130** includes a first correction unit **130_1** and a second correction unit **130_2**.

(52) In order to correct the driving capability of one of the plurality of driving sources **50** to be the correction targets, that is, the second driving source **50_2** in the example described above, the first correction unit **130_1** corrects the driving amount calculated by the calculation unit **120** based on the detection position of the lens **30** and the tilt of the lens **30**. As an example, the first correction unit **130_1** corrects the driving current corresponding to the driving amount calculated by the calculation unit **120** using, for example, the following expression “corrected driving current=driving current×coefficient 1+driving current×detection position×coefficient 2”. The first correction unit **130_1** outputs the correction driving amount obtained by correcting the driving amount to the driving unit **140** corresponding to one of the plurality of driving sources **50** to be the correction targets, that is, the second driving unit **140_2** in the example described above.

(53) In order to correct the driving capability of the other one of the plurality of driving sources **50** to be the correction targets, that is, the third driving source **50_3** in the example described above, the second correction unit **130_2** corrects the driving amount calculated by the calculation unit **120** based on the detection position of the lens **30** and the tilt of the lens **30**. As an example, when a predetermined third coefficient is set as a “coefficient 3” and a predetermined fourth coefficient is set as a “coefficient 4”, the second correction unit **130_2** corrects the driving current corresponding to the driving amount calculated by the calculation unit **120** using, for example, the following expression “corrected driving current=driving current×coefficient 3+driving current×detection position×coefficient 4”. Herein, similar to the “coefficient 1” and the “coefficient 2”, at least one of the “coefficient 3” or the “coefficient 4” may be predefined such that it becomes a numerical value corresponding to the magnitude of the tilt of the lens **30**. The second correction unit **130_2** outputs

the correction driving amount obtained by correcting the driving amount to the driving unit **140** corresponding to the other one of the plurality of driving sources **50** to be the correction targets, that is, the third driving unit **140_3** in the example described above.

(54) Also when the lens **30** is moved using four or more driving sources, sub-blocks can similarly be added to the correction unit **130** and the plurality of driving units **50**. In this manner, using one of the plurality of driving sources **50** (for example, the first driving source **50_1**) as a reference, the driving apparatus **100** according to the third embodiment corrects the driving capabilities of the plurality of driving sources **50** to be the correction targets (for example, the second driving source **50_2** and the third driving source **50_3**). Thus, according to the driving apparatus **100** of the third embodiment, also when the lens **30** is moved using three or more driving sources, the tilt of the lens **30** due to an imbalance of the driving capabilities among the plurality of driving sources **50** can be corrected according to the position of the lens **30** that has actually been detected.

(55) FIG. 5 shows an example of a block diagram of the camera module **10** that may include the driving apparatus **100** according to a fourth embodiment. In the present figure, components having the same functions and configurations as those of FIG. 4 are denoted by the same reference numerals, and descriptions thereof will be omitted below except for different points. In the third embodiment, the case where the driving apparatus **100** executes the tilt correction processing based on the detection position of the lens **30** and the tilt of the lens **30** has been described as an example. In the fourth embodiment, similar to the second embodiment, the driving apparatus **100** executes the tilt correction processing based on the target position of the lens **30** and the tilt of the lens **30**.

(56) In order to correct the driving capability of one of the plurality of driving sources **50** to be the correction targets, that is, the second driving source **50_2** in the example described above, the first correction unit **130_1** corrects the driving amount calculated by the calculation unit **120** based on the target position of the lens **30** and the tilt of the lens **30**. As an example, the first correction unit **130_1** corrects the driving current corresponding to the driving amount calculated by the calculation unit **120** using, for example, the following expression “corrected driving current=driving current×coefficient 1+driving current×target position×coefficient 2”.

(57) In order to correct the driving capability of the other one of the plurality of driving sources **50** to be the correction targets, that is, the third driving source **50_3** in the example described above, the second correction unit **130_2** corrects the driving amount calculated by the calculation unit **120** based on the target position of the lens **30** and the tilt of the lens **30**. As an example, the second correction unit **130_2** corrects the driving current corresponding to the driving amount calculated by the calculation unit **120** using, for example, the following expression “corrected driving current=driving current×coefficient 3+driving current×target position×coefficient 4”.

(58) In this manner, the driving apparatus **100** according to the fourth embodiment corrects the driving amount based on the function of the target position instead of the detection position. Thus, according to the driving apparatus **100** of the fourth embodiment, also when the lens **30** is moved using three or more driving sources, the tilt of the lens **30** due to the imbalance of the driving capabilities among the plurality of driving sources **50** can be corrected according to the target position of the lens **30** designated by the processor **70** instead of the position of the lens **30** that has actually been detected.

(59) FIG. 6 shows an example of a block diagram of the correction unit **130** in the driving apparatus **100** according to a fifth embodiment. In the embodiments described above, the case where at least one of the detection position or target position of the lens **30** is definitely used for correcting the driving amount has been described as an example. However, depending on a structure of the actuator (plurality of driving sources **50**) or the like, a case of not using the detection position or target position of the lens for the correction calculation may be preferable. Therefore, in the fifth embodiment, the correction unit **130** is configured to be capable of switching whether to use either the detection position or the target position for correcting the driving amount. The correction unit **130** includes a switching unit **135**.

(60) The switching unit **135** switches a correction method for correcting the driving amount. For example, as a correction method 1, the correction unit **130** assumes that the driving amount can be corrected using the following expression “corrected driving current=driving current×coefficient 1+coefficient 2”. In addition, as a correction method 2, the correction unit **130** assumes that the driving amount can be corrected using the following expression “corrected driving current=driving current×coefficient 1+driving current×detection position or target position×coefficient 2”.

(61) In such a case, the switching unit **135** switches the correction method for correcting the driving amount between the correction method 1 and the correction method 2 according to the structure of the actuator or the like. In this manner, in the fifth embodiment, the correction unit **130** may include the switching unit **135** that makes, according to the characteristics of the first driving source **50_1** and the second driving source **50_2**, a switch between the correction that uses either the detection position or the target position (for example, the correction method 2, that is, the correction that uses the expression “corrected driving current=driving current×coefficient 1+driving current×detection position or target position×coefficient 2”) and the correction that does not use the detection position and the target position (for example, the correction method 1, that is, the correction that uses the expression “corrected driving current=driving current×coefficient 1+coefficient 2”). Thus, according to the driving apparatus **100** of the fifth embodiment, an optimum method can be selected according to the characteristics of the actuator to thus correct the driving amount.

(62) FIG. 7 shows an example of a block diagram of the camera module **10** that may include the driving apparatus **100** according to a sixth embodiment. In the present figure, components having the same functions and configurations as those of FIG. 1 are denoted by the same reference numerals, and descriptions thereof will be omitted below except for different points. In the embodiments described above, the case where information representing the tilt of the optical element is acquired in advance has been described as an example. In the sixth embodiment, the tilt of the optical element is detected every time during an actual operation.

(63) In the sixth embodiment, the camera module **10** further includes a tilt detector **60**. In addition, the processor **70** is communicably connected to the tilt detector **60** via, for example, serial communication, parallel communication, a network, wireless communication, or the like.

(64) It is to be noted that although the present figure shows an example of a case where the tilt detector **60** is provided outside the driving apparatus **100**, the tilt detector **60** may alternatively be built-in in the driving apparatus **100**.

(65) The tilt detector **60** detects the tilt of the optical element (lens **30**). The tilt detector **60** supplies information representing the detected tilt of the lens to the processor **70**.

(66) In the sixth embodiment, the correction unit **130** acquires the information representing the tilt of the lens **30** detected by the tilt detector **60** from the processor **70**. Then, when a predetermined coefficient is set as a “coefficient 1”, the correction unit **130** may correct the driving current using the following expression “corrected driving current=driving current×coefficient 1+tilt information×F (detection position)” which directly uses the tilt information. Herein, F (detection position) is a value that varies depending on the detection position. In this manner, for example, the correction unit **130** may perform feedback so that the tilt of the lens that has been detected during the actual operation becomes equal to or smaller than a predetermined threshold (preferably zero), and thus perform the tilt correction of the lens **30**.

(67) While the embodiments of the present invention have been described, the technical scope of the present invention is not limited to the above described embodiments. It is apparent to persons skilled in the art that various alterations and improvements can be added to the above described embodiments. It is also apparent from the description of the claims that the embodiments to which such alterations or improvements are made can be included in the technical scope of the present invention.

(68) The operations, procedures, steps, and stages of each process performed by an apparatus,

system, program, and method shown in the claims, specification, or drawings can be performed in any order as long as the order is not indicated by “prior to,” “before,” or the like and as long as the output from a previous process is not used in a later process. Even if the process flow is described using phrases such as “first” or “next” in the claims, specification, or drawings, it does not necessarily mean that the process must be performed in this order.

EXPLANATION OF REFERENCES

(69) **10**: camera module; **20**: object; **30**: lens; **40**: magnet; **40_1**: first magnet; **40_2**: second magnet; **50**: plurality of driving sources; **50_1**: first driving source; **50_2**: second driving source; **50_3**: third driving source; **60**: tilt detector; **70**: processor; **100**: driving apparatus; **110**: position sensor; **120**: calculation unit; **130**: correction unit; **130_1**: first correction unit; **130_2**: second correction unit; **135**: switching unit; **140**: plurality of driving units; **140_1**: first driving unit; **140_2**: second driving unit; **140_3**: third driving unit.

Claims

1. A driving apparatus, comprising: a calculation unit configured to calculate a driving amount of an optical element based on a detected position of the optical element and a target position of the optical element; a correction unit to which the driving amount, the detected position, and the target position are input, the correction unit being configured to generate a corrected driving amount obtained by correcting the driving amount based on the driving amount, a parameter corresponding to a tilt of the optical element, and either the detected position or the target position, and output the driving amount and the corrected driving amount to autofocus or zoom the optical element; a first driving unit configured to apply a first driving force corresponding to the driving amount received from the correction unit to a first driving source out of a plurality of driving sources for moving the optical element in a direction of an original optical axis when the optical element is not tilted, wherein the first driving source is for moving a first edge of the optical element a first moved amount in the original optical axis direction; and a second driving unit configured to apply a second driving force corresponding to the corrected driving amount received from the correction unit to a second driving source out of the plurality of driving sources, wherein the second driving source is for moving a second edge of the optical element, on an opposite side of the optical element to the first edge, a second moved amount in the original optical axis direction.
2. The driving apparatus according to claim 1, wherein the correction unit is configured to correct the driving amount based on a function of the detected position.
3. The driving apparatus according to claim 2, wherein when a predetermined first coefficient is set as a “coefficient 1” and a predetermined second coefficient is set as a “coefficient 2”, the correction unit is configured to correct a driving current corresponding to the driving amount calculated by the calculation unit using an expression “corrected driving current=driving current x coefficient 1+driving current×detected position×coefficient 2”.
4. The driving apparatus according to claim 1, wherein the correction unit is configured to correct the driving amount using a correction table predefined based on the detected position and the driving amount.
5. The driving apparatus according to claim 1, wherein the correction unit is configured to correct the driving amount based on a function of the target position.
6. The driving apparatus according to claim 5, wherein when a predetermined first coefficient is set as a “coefficient 1” and a predetermined second coefficient is set as a “coefficient 2”, the correction unit is configured to correct a driving current corresponding to the driving amount calculated by the calculation unit using an expression “corrected driving current=driving current x coefficient 1+driving current x target position x coefficient 2”.
7. The driving apparatus according to claim 1, wherein the correction unit is configured to correct the driving amount using a correction table predefined based on the target position and the driving

amount.

8. The driving apparatus according to claim 1, wherein the correction unit is configured to be capable of switching whether to use either the detected position or the target position for correcting the driving amount.

9. The driving apparatus according to claim 2, wherein the correction unit is configured to be capable of switching whether to use either the detected position or the target position for correcting the driving amount.

10. The driving apparatus according to claim 3, wherein the correction unit is configured to be capable of switching whether to use either the detected position or the target position for correcting the driving amount.

11. The driving apparatus according to claim 4, wherein the correction unit is configured to be capable of switching whether to use either the detected position or the target position for correcting the driving amount.

12. The driving apparatus according to claim 8, wherein the correction unit includes a switching unit configured to make, according to characteristics of the first driving source and the second driving source, a switch between a correction that uses either the detected position or the target position and a correction that does not use the detected position and the target position.

13. The driving apparatus according to claim 1, wherein the correction unit is configured to select, based on the parameter corresponding to the tilt of the optical element, which of the plurality of driving sources is to be used as a reference for correcting the driving amount.

14. The driving apparatus according to claim 2, wherein the correction unit is configured to select, based on the parameter corresponding to the tilt of the optical element, which of the plurality of driving sources is to be used as a reference for correcting the driving amount.

15. The driving apparatus according to claim 3, wherein the correction unit is configured to select, based on the parameter corresponding to the tilt of the optical element, which of the plurality of driving sources is to be used as a reference for correcting the driving amount.

16. The driving apparatus according to claim 13, wherein the parameter corresponding to the tilt indicates a tilt of the optical element that is detected when a maximum current is caused to flow into each of the plurality of driving sources.

17. The driving apparatus according to claim 1, wherein the calculation unit is configured to calculate the driving amount by feedback control that uses the detected position and the target position.

18. The driving apparatus according to claim 2, wherein the calculation unit is configured to calculate the driving amount by feedback control that uses the detected position and the target position.

19. The driving apparatus according to claim 3, wherein the calculation unit is configured to calculate the driving amount by feedback control that uses the detected position and the target position.

20. A driving method for driving a plurality of driving sources including a first driving source and a second driving source, comprising: detecting a tilt of an optical element; generating a parameter corresponding to the tilt of the optical element; detecting a position of the optical element and outputting a detected position; calculating a driving amount of the optical element based on the detected position of the optical element and a target position of the optical element; generating a corrected driving amount obtained by correcting the driving amount based on the driving amount, the parameter corresponding to the tilt of the optical element, and either the detected position or the target position; and based on the parameter corresponding to the tilt of the optical element, selecting the first driving source to be driven by a first driving force corresponding to the driving amount and selecting the second driving source to be driven by a second driving force corresponding to the corrected driving amount, wherein the first driving source is for moving a first edge of the optical element a first moved amount in an original optical axis direction, and wherein

the second driving source is for moving a second edge of the optical element, on an opposite side of the optical element to the first edge, a second moved amount in the original optical axis direction to autofocus or zoom the optical element.

21. The driving apparatus according to claim 1, wherein the first moved amount is different from the second moved amount.
